

for supercapacitors range from memory backup and LED flash lights, to critical applications. With an optimistic prediction of 20 per cent penetration by 2020, estimated demand in this sector will add up to over 37 million units of supercapacitors of various capacities.

Defence. High specific power and higher life cycles (up to millions of charge/discharge cycles), high efficiencies up to 98 per cent, quick charge/discharge and workability in extreme weather conditions make supercapacitors an efficient storage option for the defence sector. However, defence, being a conservative sector in nature, might not allow easy entry for any new technology.

Power. The world's population is expected to grow by two billion people by 2050, and global energy demand is expected to roughly double during the same period. Concurrently, the power sector is on the brink of a major transformation, like adoption and grid integration of solar and wind energy.

Unlike in the past, the grid must now accommodate producers and consumers with disparate usage patterns. Stability of the power grid depends on various actors working in tandem to maintain a balance between electricity supply and demand. Traditionally, electricity assets were categorised based on their function, that is, generation, transmission and distribution.

The smart grid concept has made inroads owing to the above-mentioned situation. There is demand for smart supercapacitor-based storage systems, which can balance supply and demand across segments that comprise the value chain. New control points offered by supercapacitor-based storage systems enable operators to selectively and instantly respond to fluctuations in grid inputs and outputs.

Such functionality is an essential requirement of the smart cities concept, where producers and consumers are equally informed and equipped to respond to market dynamics in real time.

The Indian government has released its first list of smart cities. In the coming years, supercapacitors will be in high demand in this sector for

smart grid applications, smart utility energy meter applications and so on.

Market challenges

High costs and technical issues (like low energy density) are two key factors that are hindering growth optimisation. While researchers are exploring alternative supercapacitor electrode materials like graphene to address the issue with low energy density, mass production and wide application will bring down the costs further.

Worldwide market players are pinning their hopes on the automotive industry, particularly the hybrid EV segment, to increase volume demand. This increasing demand will consequently trigger significant price declination.

Potential manufacturers need support to make supercapacitors, right from ensuring successful adoption of technology. A wider indigenous manufacturing base for supercapacitors will help improve the scenario for new entrants.

A major manufacturing challenge is that supercapacitor technology is still in evolution mode, with a lot of improvements lined up.

Another challenge is unavailability of indigenous raw materials, with almost everything being imported, including aluminium foils, separators, carbon and electrolytes.

Moving forward

Supercapacitors are known for being environment-friendly and efficient; for their performance, reliability, fast-charging capabilities, virtually infinite charge/discharge life cycles, being maintenance-free; and for having a wide temperature range. All these properties pay in the long run, on the environment and economic fronts, besides leading to power savings, improving power quality and contributing towards national fuel security policies.

The global economy rides on the automotive sector, and the Indian scenario is no different. Globally, the automotive sector is being geared for changing over to electric mobility. Supercapacitors are vital to its successful implementation.

Manufacturing of supercapacitors in India directly falls under the import-substitute category. Therefore

creation of indigenous supercapacitor manufacturing facilities will directly result in savings on import bills. And considering the growing level of global adoption of supercapacitors, India may have opportunities for export too. Venturing into indigenous manufacturing of supercapacitors will certainly improve India's balance of payments and add foreign exchange reserves.

Though supercapacitor technology is in the R&D stage in India, opportunity for this technology is huge considering the various application sectors. To get the technology in mainstream and start mass production, Indian market needs to have some levers that will foster the adoption of the technology here.

As the initial steps to get the supercapacitor technology in mainstream and start mass production, India can look at the following solutions:

Using supercapacitors as hybrid energy storage solution. An immediate solution for supercapacitor manufacturers or designers would be to pair these with batteries, creating a hybrid energy storage solution.

Creating demand in the market. By offering subsidies, government-industry collaborations and purchase mandates, supercapacitor technology can be promoted.

Supply-side interventions. By providing OEM incentives to promote local manufacturing and encouraging tech startups to work on advanced and emerging technologies like supercapacitor, enough interest can be generated among local manufacturers to accelerate indigenous production.

Infrastructure development. A complete supply chain is required to boost local manufacturing. This would also keep the end-product cost at the lower side, since India will not have to import raw materials or any other components.

Business models for commercialisation. To initiate commercialisation of supercapacitor technology, business models such as joint venture strategy, cross-industry consolidation and R&D outsourcing can be adopted. These business models have advantages such as faster time-to-market, risk transfer, knowledge sharing and access to expertise. **EFY**